

Online Resource 1

Mechanistic Interpretability of a Plant Genomic Foundation Model

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Supplementary Figures

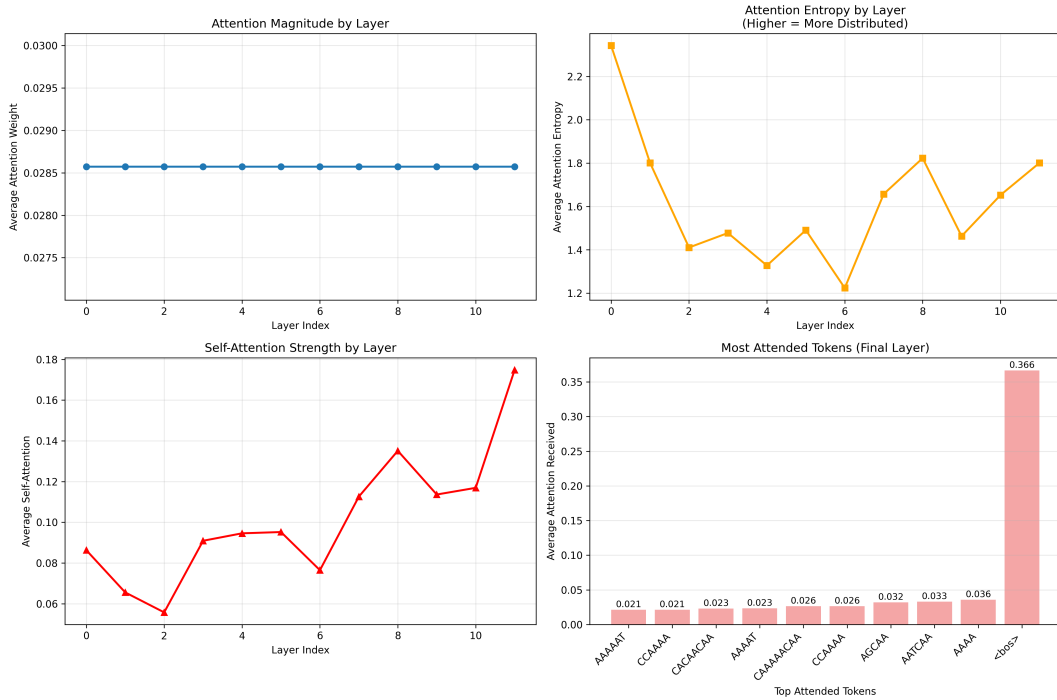


Figure 1: **Figure S1: Attention weight statistics.** Distribution of mean attention weights, entropy, and maximum attention values across all 144 attention heads in Plant-DnaGemma. The diversity of attention statistics indicates that heads employ varied strategies ranging from highly focused to broadly distributed attention.

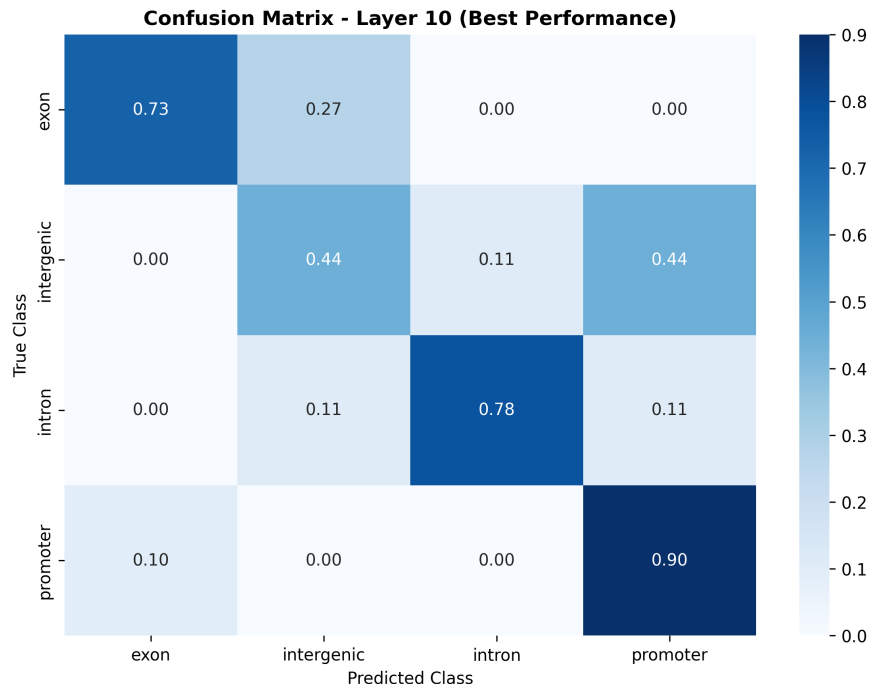


Figure 2: **Figure S2: Confusion matrix at the best probing layer.** Classification patterns across the four genomic sequence types at the best-performing layer. Off-diagonal entries reveal which sequence types are most commonly confused.

Table 1: **Table S1: k -mer baseline results for sequence type classification.** 5-fold cross-validated accuracy and macro F1 for k -mer frequency features.

Method	Features	Accuracy	Macro F1	Std
3-mer frequencies	64	0.650	—	0.045
4-mer frequencies	256	0.657	—	0.047
5-mer frequencies	1,024	0.670	—	0.059
Random model (best)	768	0.525	0.524	0.022
Trained model (best)	768	0.745	0.743	0.044

Table 2: **Table S2: Cross-species classification with baselines.** All results are 5-fold cross-validated. Chance level is 33.3% for 3-class classification.

Method	Accuracy	Std
Chance	0.333	—
5-mer frequencies	0.457	0.048
4-mer frequencies	0.507	0.027
3-mer frequencies	0.520	0.041
Model layer 8 (best)	0.590	0.048
GC-content only	0.610	0.040

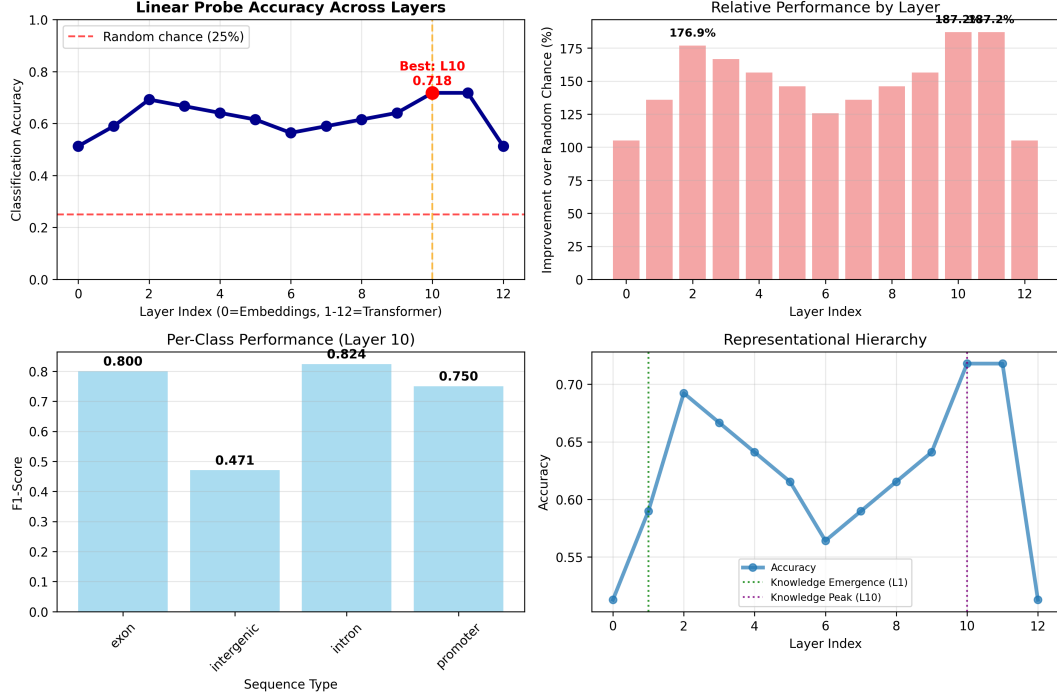


Figure 3: **Figure S3: Extended probing analysis.** Additional probing visualizations including per-class F1 scores, relative improvement over chance, and representational hierarchy annotations from the initial (130-sequence) probing analysis.

Supplementary Tables

Supplementary Methods

Synthetic sequence generation

Sequences were generated to reflect key biological properties of each genomic region type:

- **Promoters:** AT-rich background (60% AT), with TATA box (TATAAA) inserted at positions 20–35, CAAT box at positions 60–80, and optional GC box (GGGCGG) at positions 100–120.
- **Exons:** Balanced nucleotide composition (25% each), with in-frame stop codons (TAA, TAG, TGA) replaced by random non-stop codons, and optional start codon (ATG).
- **Introns:** AT-rich background (65% AT), with canonical 5' splice site (GT) at position 0, 3' splice site (AG) at the end, and an A-rich branch point region at the sequence midpoint.

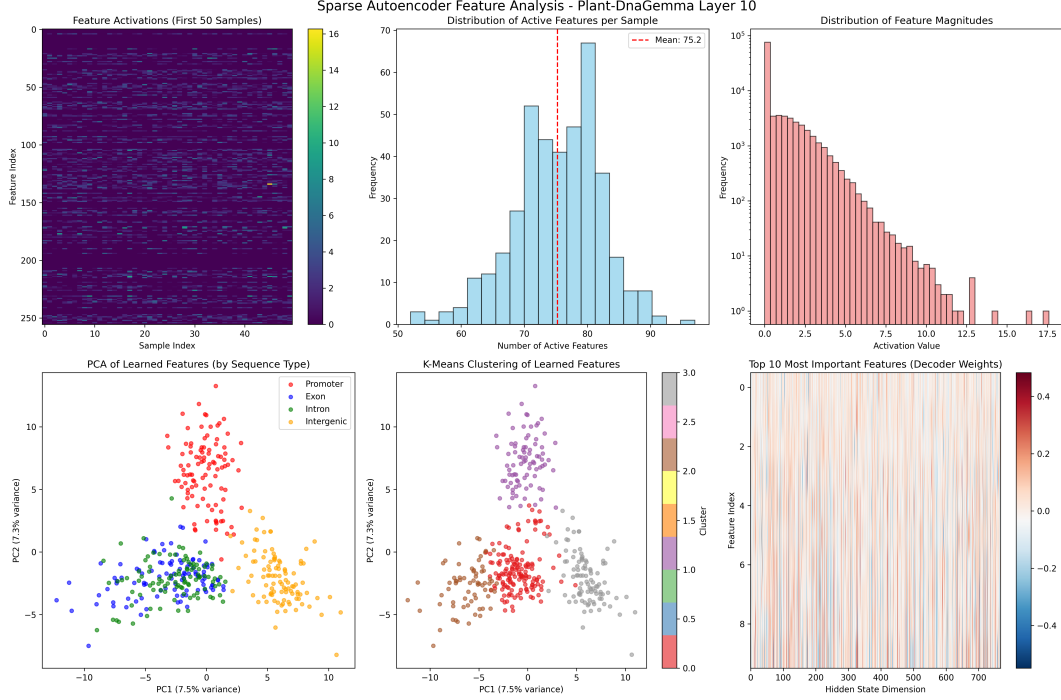


Figure 4: **Figure S4: Initial bottleneck SAE analysis.** For comparison with the overcomplete SAE in the main paper, this shows results from a bottleneck autoencoder (768→256) trained on layer-10 representations. The bottleneck architecture achieves 70.6% sparsity but with only 256 features, limiting the granularity of feature discovery.

- **Intergenic:** AT-rich background (60% AT) with optional short tandem repeats (AT, TA, CA, TG) inserted at random positions.

Cross-species sequence generation

Species-specific sequences were generated by sampling nucleotides according to species-appropriate GC content: 36% for *A. thaliana*, 44% for *O. sativa*, and 47% for *Z. mays*. These values reflect documented genome-wide GC content for each species.

Overcomplete SAE architecture details

The overcomplete sparse autoencoder uses the following architecture:

- Encoder: Linear(768, 3072) + ReLU
- Decoder: Linear(3072, 768)
- Loss: MSE reconstruction + $\lambda \cdot \ell_1$ penalty on encoder activations ($\lambda = 5 \times 10^{-4}$)

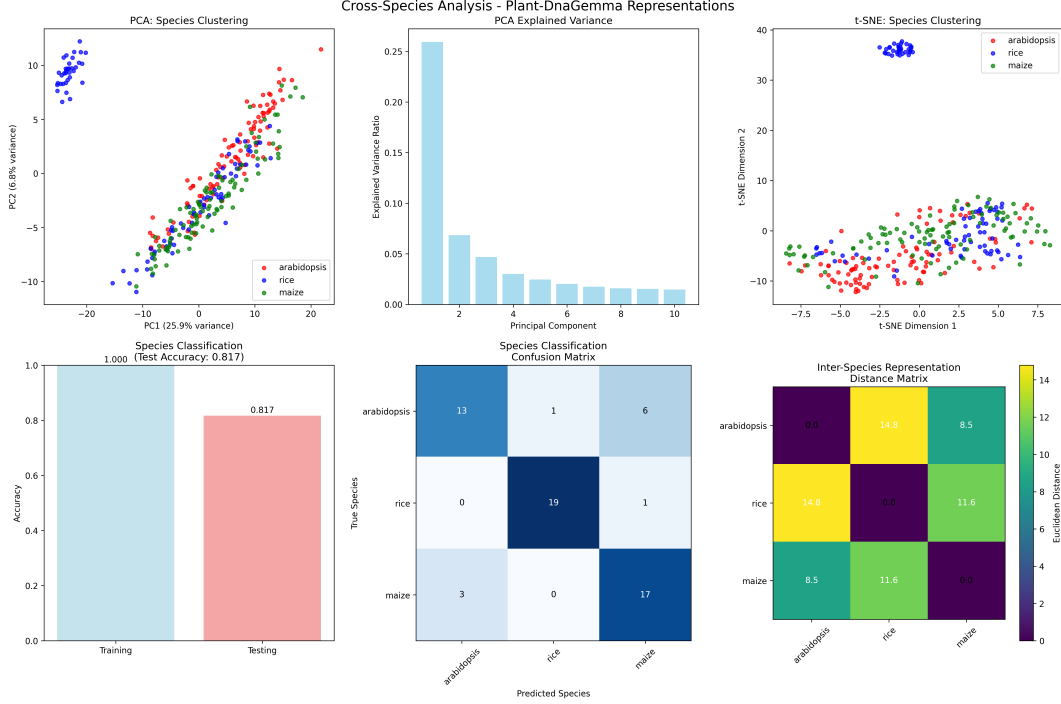


Figure 5: **Figure S5: Initial cross-species analysis.** PCA and t-SNE projections of layer-10 representations for *Arabidopsis*, rice, and maize sequences, with species classification and inter-species distance analysis. Note that these results were obtained from a single train/test split; the main paper reports 5-fold cross-validated results with baseline comparisons.

- Optimizer: Adam with learning rate 10^{-3}
- Input preprocessing: Per-feature standardization (zero mean, unit variance)
- Training: 200 epochs, batch size 64

This follows the overcomplete dictionary learning paradigm of Bricken et al (2023), where the dictionary size (3,072) exceeds the input dimension (768) by a factor of $4\times$, allowing features to be represented as sparse combinations of dictionary elements rather than compressed encodings.

References

Bricken T, Templeton A, Batson J, et al (2023) Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread